

# SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

## COURSE OUTCOME COVERED

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

### Annexure A

### SHORT ANSWER TEST

#### Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

#### Code of Conduct:

B. Chaitanya Prakash Reddy (192425255) .

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

*Chaitanya*

Signature of the student:

| Criteria                  | Weightage | Q1 10marks | Q2 10marks | Q3 10marks | Q4 10marks |
|---------------------------|-----------|------------|------------|------------|------------|
| Understanding of Concepts | 25%       | L          | L          | L          | L          |
| Explanation               | 25%       | L          | L          | L          | L          |
| Application               | 20%       | L          | L          | L          | L          |
| Organization              | 15%       |            |            |            |            |
| Accuracy                  | 10%       |            |            |            |            |
| Clarity                   | 5%        |            |            |            |            |

Total Marks *26* /40

Signature of the Faculty *[Signature]*

## FTP Data Transfer

- Total Data = 3GB = 3000MB.
- Size of each request = 3MB.

Number of requests:

$$= 3000 \div 3 = 1000 \text{ requests.}$$

- Throughput = 120Mbps.
- Total data in bits =  $3 \times 1000 \times 8 = 24,000 \text{ Mb.}$

Time = Total Data / Throughput.

$$= 24,000 / 120$$

$$= 200 \text{ Sec.}$$

\* Total Request Generated = 1000

\* Total Time = 200 Sec.

## Application layer services:

- Provides user-friendly service.
- Ensures reliable file transfer.
- Handles authentication and access control.

## 2. End-to-End Delay Calculation.

Application Layer 4ms

Transport Layer 3ms

Network Layer 5ms

Data link layer 2ms

Physical layer 1ms

Transmission layer 12ms.

Propagation layer 18ms.

Total Delay:

$$4 + 3 + 5 + 2 + 1 + 12 + 18 = 45 \text{ ms.}$$

• Total = 45ms.

Application layer: Provides services to application.

Transport layer: ensures reliable delivery.

Network layer: Determines routing Path.

Data link layer: Handles framing and error detection.

Physical Layer: Transmits bits over medium.

## 9. Frame overhead and Efficiency.

- useful data =  $80 \text{ MB} = 80 \times 10^6$
- Frame Size = 1600
- overhead per Frame = 80 bytes.
- useful =  $1600 - 80 = 1520$  bytes.

$$\begin{aligned}\text{No. of frames} &= 80,000,000 \div 1520 \\ &= 52,632 \text{ frames.}\end{aligned}$$

$$\text{Total overhead} = 52632 \times 80 = 4.21 \text{ MB.}$$

Total Data Sent:

$$80,000,000 + 4,210,560 = 84,210,560$$

$$\text{No. of frames} = 52,632.$$

$$\text{Total overhead} = 4.21 \text{ MB.}$$

$$\text{Efficiency} = 95\%.$$

- \* Increases bandwidth usage.
- \* Reduces transmission efficiency.

10.

## Hospital Medical Record.

- original File Size = 240 MB.

- compression = 30 %.

$$= 240 \times 0.70 = 168 \text{ MB.}$$

- segment size = 1400 bytes.

- Data link overhead = 60 bytes.

Compressed data:

$$168 \text{ MB} = 168,000,000 \text{ bytes.}$$

No. of segments:

$$= 168,000,000 \div 1400 = 120,000 \text{ segments.}$$

Total protocol:

$$= 120,000 \times 60 = 7,200,000 = 7.2 \text{ MB}$$

=> Compressed file = 168 MB

Total segments = 120,000

Total protocol = 7.2 MB.

Final data Transmitted = 175.2 MB.